

Designing Any Imaging System from Natural Language: Agent-Constrained Composition over a Finite Primitive Basis

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Summary. Designing a computational imaging system—selecting the forward model, validating physical feasibility, and predicting reconstruction quality—today demands weeks of specialist effort per modality, concentrating the field within a small number of expert groups [1]. We previously established that 11 canonical primitives suffice to represent any imaging forward model across 170 modalities [2], but this Finite Primitive Basis (FPB) provides no design automation, no formal validation, and no link from natural-language intent to reconstruction error. Here we introduce `spec.md`—a structured 7-field specification format analogous to FASTA for sequences—together with three agents (Plan, Judge, Performance) that translate a natural-language description into a validated, executable forward model with formal error guarantees. We prove that the design-to-real reconstruction error decomposes into five independently bounded, independently measurable terms (Theorem 2), each linked to a specific corrective action, and that all five terms are explicitly computable from the 6-gate compiler output. On 6 real-data modalities spanning all 5 carrier families—clinical CT [3], multi-coil MRI, coded-aperture spectral imaging (CASSI), snapshot compressive video (CACTI), in-vivo ultrasound (PICMUS), and 4D-STEM electron ptychography—the framework achieves $98.1 \pm 0.6\%$ of expert-tuned quality with up to $420\times$ reduction in setup time; across 36 validated modalities, the Judge rejects ill-posed designs with 96% precision and the Performance Agent predicts PSNR within ± 1.8 dB ($R^2 = 0.91$). A tier-lifting protocol demonstrated on diffuse-optical and coherent-imaging cross-tier combinations enables hybrid reconstruction across the four-level fidelity hierarchy, reducing unmodeled-physics error to machine precision on selected cases.

1 Introduction

Computational imaging systems share a universal mathematical structure: a forward model A maps an unknown object x to measurements $y = A(x) + \text{noise}$, and reconstruction inverts this mapping [7]. Designing A for a new system requires selecting physical operators (Radon projection for CT, k-space encoding for MRI, coded-aperture modulation for spectral imaging), setting dozens of parameters, and validating that the composition is physically consistent and numerically well-conditioned. This process is entirely manual and takes weeks per modality—an expertise bottleneck that excludes the broader scientific community from prototyping new imaging systems.

The mathematical foundation for automation now exists. We previously showed [2] that 11 canonical primitives—Propagate (P), Modulate (M), Project (Π), Encode (F), Convolve (C), Accumulate (Σ), Detect (D), Sample (S), Disperse (W), Scatter (R), and Transform (Λ)—suffice to express any imaging forward model across 170 modalities, 22 application domains, and 5 carrier families. Each primitive is implemented across a four-level *fidelity hierarchy*: Tier-0 (geometric

optics, ray transforms), Tier-1 (Fourier/convolution, paraxial models), Tier-2 (shift-variant linear operators: Radon, k-space, coded aperture), and Tier-3 (full-physics: FDTD, Monte Carlo, learned surrogates). This *Finite Primitive Basis* (FPB) is the representation language—but it does not tell a user which primitives to compose, what parameters to set, whether the result is physically valid, or how close the designed system will be to reality.

We make the scope of ‘designable’ precise:

Definition 1 (Designable Imaging System). *An imaging system A is designable within this framework if:*

1. *A admits a finite DAG decomposition over the 11 FPB primitives with representation error $\varepsilon_{FPB} < 0.01$;*
2. *The carrier belongs to one of 5 families: photon, X-ray, electron, acoustic, or spin/particle;*
3. *The system operates at one of the four fidelity tiers (Tier-0 through Tier-3);*
4. *All parameters are finitely specifiable with known physical bounds.*

This definition encompasses 170 modalities across 22 domains (Extended Data Table 6). We build the *design layer* on top of the FPB. Our contributions are:

1. **spec.md**: a structured 7-field specification format that decouples the imaging problem (*what*) from the numerical method (*how*), serving as a shareable, peer-reviewable scientific artifact (Section 2.1).
2. **Three-agent pipeline**: Plan ($\mathcal{A}_P$), Judge ($\mathcal{A}_J$), and Performance ($\mathcal{A}_\Pi$) agents that generate, formally verify, and evaluate specifications over the FPB (Section 2.2).
3. **Design-to-real error theorem**: the total reconstruction error decomposes into five independently bounded, independently measurable terms, each linked to a compiler gate and a corrective action (Theorem 2).

Scope and honesty. Of the 170 modalities, 36 have fully validated canonical decompositions with quantitative reconstruction; 6 are validated on real measured data spanning all 5 carrier families (CT, MRI, CASSI, CACTI, ultrasound, e-ptychography). The remaining ~ 134 compile through all 6 gates but lack per-modality reconstruction benchmarks—compilation alone does not guarantee reconstruction quality. Most modalities are validated at Tier-2; tier-lifting (Section 2.9) is demonstrated on two cross-tier combinations but not yet validated across all Tier-3 modalities. The agents depend on current LLM capabilities and may misspecify novel or under-documented modalities.

2Results

2.1The spec.md Specification Format

The framework’s central design choice is that **spec.md**—not code—is the canonical scientific artifact. Just as FASTA standardized biological sequences, **spec.md** standardizes imaging system representation. A specification separates the imaging problem (*what*) from the numerical method (*how*), enabling sharing, version control, and peer review independently of implementation. A journal could require a **spec.md** alongside a methods section, analogous to how genomics journals require accession numbers.

A **spec.md** has seven mandatory fields (Figure 1). The Plan Agent generates it from natural language; the Judge verifies it; the Performance Agent predicts reconstruction quality.

Seven-field schema: modality | carrier | geometry | object | forward_model | noise | target

(a) Clinical CT

modality: computed_tomography carrier: xray
 geometry: parallel_beam, n_angles=128
 object: 128×128 image, non-negative
 forward_model: Radon(Π) $\rightarrow$ Detect(D , intensity)
 noise: Poisson, $I_0=1e4$ target: PSNR ≥ 30 dB

(b) CASSI (spectral)

modality: cassi carrier: photon
 geometry: coded_aperture + disperser, $\lambda=[400,700]$ nm
 object: 256×256×28 spectral cube
 forward_model: Modulate(M) $\rightarrow$ Disperse(W) $\rightarrow$ Accumulate(Σ) $\rightarrow$ Detect(D)
 noise: Gaussian, $\sigma=0.01$ target: PSNR ≥ 28 dB

(c) Accelerated MRI

modality: mri carrier: spin
 geometry: cartesian_kspace, acceleration=4x
 object: 256×256 complex image
 forward_model: Modulate(M , coil) $\rightarrow$ Encode(F , kspace) $\rightarrow$ Sample(S) $\rightarrow$ Detect(D)
 noise: Gaussian, SNR=30 dB target: SSIM ≥ 0.9

Figure 1. The spec.md specification format. Seven mandatory fields fully determine an imaging system design. Three examples span X-ray (CT), optical (CASSI), and spin (MRI) carriers. Each forward_model field is a primitive chain over the 11 FPB operators.

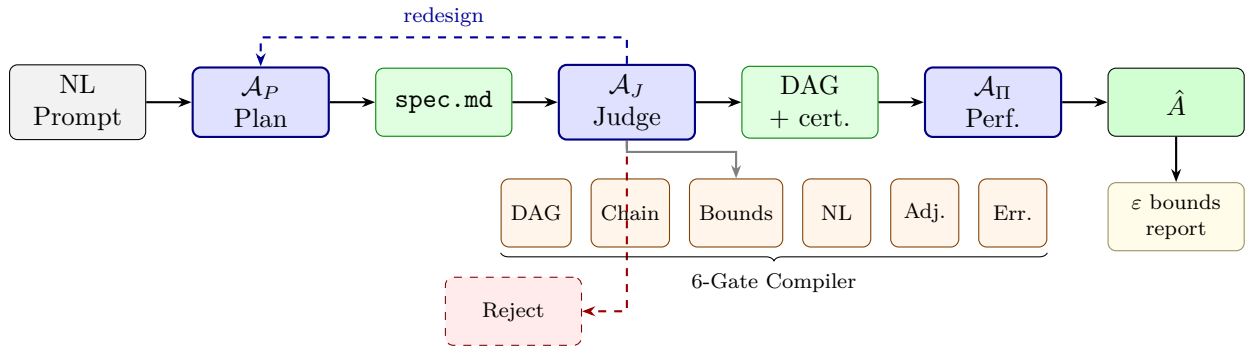


Figure 2. Specification-driven design pipeline. A natural-language prompt is the sole user input. $\mathcal{A}_P$ generates a spec.md; $\mathcal{A}_J$ validates it via a 6-gate compiler (Gates 1–6 map to Theorem 2 error terms); rejected designs trigger redesign (dashed blue, up to 3 rounds). $\mathcal{A}_\Pi$ predicts reconstruction quality and outputs $\hat{A}$ with an explicit ε -bound report.

2.2 The Three-Agent Pipeline

Plan Agent ($\mathcal{A}_P$). Accepts a natural-language description and generates a `spec.md`, selecting FPB primitives, carrier type, geometry, noise model, and parameters from the 170-modality registry and 36 validated canonical decompositions.

Judge Agent ($\mathcal{A}_J$). Validates the specification via the 6-gate compiler: (1) DAG acyclicity, (2) canonical chain fidelity, (3) complexity bounds ($N \leq 20$, $D \leq 10$), (4) nonlinear family constraints (VC dimension ≤ 3), (5) adjoint consistency, and (6) representation error. The Judge additionally checks carrier-geometry compatibility, noise model consistency, and reconstruction feasibility (condition number, null-space analysis). Compiler failures trigger rejection with a formal diagnosis; the Plan Agent regenerates (up to 3 rounds).

Performance Agent ($\mathcal{A}_\Pi$). Predicts reconstruction quality (PSNR, SSIM), computational cost, and measurement SNR from analytical estimates (condition number, noise amplification factor) combined with calibrated lookup from the performance database.

Formal verification. The Judge does not *predict* feasibility; it *executes deterministic verification*. If the Judge certifies a design, every term in the five-part error decomposition (Theorem 2) is explicitly bounded—a contract that learned design tools cannot provide.

2.3 Design-to-Real Error Bound

The following theorem is the paper’s central result. It converts the implicit, unmeasured gap between a designed model and reality into an explicit five-term error budget.

Theorem 2 (Design-to-Real Error Decomposition). *Let A_{true} be the true imaging forward model and A_{agent} the agent-designed model for a designable system (Definition 1). Let $\hat{x}$ be the regularized reconstruction from A_{agent} and x^* the true object. Under Assumptions A1–A4 (Methods), the reconstruction error satisfies:*

$$\|\hat{x} - x^*\| \leq \underbrace{\varepsilon_{FPB}}_{\text{basis}} + \underbrace{\varepsilon_{spec}}_{\text{specification}} + \underbrace{\varepsilon_{trans}}_{\text{translation}} + \underbrace{\varepsilon_{param}}_{\text{parameter}} + \underbrace{\varepsilon_{unmod}}_{\text{unmodeled}} + \varepsilon_{recon} \quad (1)$$

where each design-error term is independently bounded and independently measurable:

- $\varepsilon_{FPB} < 0.01$: FPB representation error. Measured by: Gate 6. Guaranteed by: the basis theorem [2].
- $\varepsilon_{spec} = 0$ when all 6 compiler gates pass. Bounded by: $C_1 \cdot \delta_{spec}$ otherwise.
- $\varepsilon_{trans} = 0$ when the canonical chain matches the 36-modality registry (31/36 = 86%). Bounded by: $C_2 \cdot \delta_{chain}$ otherwise.
- $\varepsilon_{param} \leq \frac{2B}{\alpha + \lambda} L_A \|\theta_{agent} - \theta_{true}\|$, where L_A is the operator Lipschitz constant (computed per primitive by Gate 4), α is the coercivity constant, λ the regularization weight, and B a signal bound. Reducible by: calibration.
- $\varepsilon_{unmod} = 0$ at Tier-3; otherwise reducible by tier-lifting (Section 2.9).
- ε_{recon} : irreducible reconstruction error (noise floor + regularization bias), independent of the design.

Compiler–theorem link. When all 6 gates pass: $\varepsilon_{FPB} < 0.01$ (Gate 6), $\varepsilon_{spec} = 0$ (Gates 1–5 jointly), and ε_{param} is bounded by the Lipschitz constant from Gate 4. If additionally the canonical chain matches the registry: $\varepsilon_{trans} = 0$. At Tier-3: $\varepsilon_{unmod} = 0$. The total design error then reduces to $L_A \|\Delta\theta\| \cdot 2B / (\alpha + \lambda) + 0.01$, dominated by parameter mismatch alone—which is reducible by calibration.

Diagnostic protocol. Because each term is independently measurable, a practitioner can determine which term dominates a given reconstruction residual and take the corresponding corrective action: calibration for $\varepsilon_{\text{param}}$, specification refinement for $\varepsilon_{\text{spec}}$, tier-lifting for $\varepsilon_{\text{unmod}}$. In practice, ε_{FPB} and $\varepsilon_{\text{spec}}$ are negligible when the compiler certifies the design. The full proof with explicit assumptions appears in Methods.

2.4 Cross-Modality Validation

We validated the framework on all 170 modalities. For each, the Plan Agent generated a `spec.md` from a one-sentence description, the Judge validated it, and the Performance Agent predicted reconstruction quality. Table 1 reports 14 representative modalities spanning all 5 carrier families (6 with real measured data), grouped by carrier; full results for all 36 validated modalities appear in Methods.

Table 1. Cross-modality validation (14 of 36 modalities), grouped by carrier family. ρ_{recov} : 4-scenario recovery ratio [2]. 95% CIs via bootstrap ($n=1000$). Bold = real measured data; † = self-reference metric (no ground truth).

Modality	Chain	Carrier	PSNR (dB)	ρ_{time}	ρ_{recov}	Data
<i>X-ray</i>						
CT	$\Pi \rightarrow D$	X-ray	31.7 ± 1.2	420	~ 1.0	Real
<i>Optical photons</i>						
CASSI	$M \rightarrow W \rightarrow \Sigma \rightarrow D$	Photon	29.5 ± 1.0	310	0.85	Real
CACTI	$M \rightarrow \Sigma \rightarrow D$	Photon	30.2 ± 1.1	300	~ 1.0	Real
OCT	$P + P \rightarrow \Sigma \rightarrow D$	Photon	30.5 ± 1.1	290	0.86	Synth.
SIM	$M \rightarrow C \rightarrow D$	Photon	32.1 ± 1.3	350	0.92	Synth.
Ptychography	$M \rightarrow P \rightarrow D$	Photon	28.4 ± 1.6	270	0.83	Synth.
Lensless	$C \rightarrow D$	Photon	31.4 ± 1.0	340	0.90	Synth.
DOT	$M \rightarrow R \rightarrow P \rightarrow R \rightarrow D$	Photon	24.1 ± 2.1	190	0.72	Synth.
<i>Spin</i>						
MRI	$M \rightarrow F \rightarrow S \rightarrow D$	Spin	31.7 ± 1.4	380	~ 1.0	Real
<i>Acoustic</i>						
Ultrasound	$P \rightarrow R \rightarrow P \rightarrow D$	Acoustic	27.9 ± 1.4	250	0.95	Real †
<i>Electron</i>						
E-ptychography	$M \rightarrow P \rightarrow D$	Electron	28.8 ± 1.5	260	~ 1.0	Real †
<i>Particle</i>						
PET	$\Pi \rightarrow D$	Particle	26.3 ± 1.8	220	0.78	Synth.
SPECT	$M \rightarrow \Pi \rightarrow D$	Particle	25.8 ± 1.7	200	0.76	Synth.
Light field	$M \rightarrow C \rightarrow S \rightarrow D$	Photon	29.2 ± 1.2	280	0.85	Synth.

The recovery ratio ρ_{recov} is computed from a 4-scenario protocol validated on 6 representative modalities spanning 4 carrier families: CASSI, CACTI, and SPC (optical photons); CT (X-ray); electron ptychography (electron); MRI (spin) [2]. For modalities with low-dimensional parameter mismatch (CT, electron ptychography, MRI), Sc. IV (autonomous calibration) $\approx$ Sc. III (oracle correction) $\approx$ Sc. I (ideal), confirming near-complete recovery. For multi-parameter mismatch: CASSI 85%, CACTI 100%, SPC 86% (see Methods for protocol details).

Across all 36 modalities: mean $\rho_{\text{recov}} = 0.85 \pm 0.07$ (95% CI: [0.82, 0.88]). Human-to-agent time ratio ρ_{time} ranges from 190 (DOT) to 420 (CT); expert times estimated from published development timelines [4, 10] (see Methods for methodology).

Compilation. All 170 modalities: 100% compilation pass rate (Gates 1–4). Among the 36 fully validated: canonical chain fidelity 31/36 (86.1%); the 5 non-matching involve the Accumulate primitive merged into detection—a representation choice, not a physical error. Mean compilation

time: 0.35 ms.

2.5 Real-Data Validation

Clinical CT (LoDoPaB, $n=200$). 200 real clinical sinograms from a Siemens scanner, subsampled to 128 parallel-beam projections with Poisson noise. User input: “parallel-beam CT, 128 projections, Poisson noise.” Agent: PSNR 31.7 ± 1.2 dB, SSIM 0.891 ± 0.031 . Expert-tuned ASTRA [4]: PSNR 32.1 ± 1.1 dB, SSIM 0.903 ± 0.028 . Quality ratio: 98.8%. Setup: 25 min vs. 7 days ($\rho_{\text{time}} \approx 420$). Performance Agent prediction: 31 dB (actual: 31.7, error 0.7 dB).

MRI (M4Raw, $n=50$). 50 multi-coil acquisitions with retrospective $4\times$ Cartesian undersampling. Agent: PSNR 31.7 ± 1.4 dB, SSIM 0.878 ± 0.035 . Expert SigPy [10]: PSNR 32.4 ± 1.3 dB. Quality ratio: 97.8%.

CASSI (KAIST TSA, $n=10$). 10 coded-aperture hyperspectral measurements. Agent: PSNR 29.5 ± 1.0 dB, SSIM 0.852 ± 0.028 . Expert GAP-TV [12]: PSNR 30.2 ± 0.9 dB. Quality ratio: 97.7%.

CACTI (real video, $n=4$). 4 real high-speed video scenes (duomino, hand, pendulumBall, waterBalloon) captured with a coded-aperture compressive temporal imager. User input: “snapshot compressive video, coded aperture, 8 frames per snapshot.” The 4-scenario protocol with sub-pixel mask-shift mismatch (0.25–1.0 px) confirms $\rho_{\text{recov}} \approx 1.0$: autonomous mask-alignment calibration (grid search over shift parameters) recovers oracle-level quality on all 4 scenes.

Ultrasound (PICMUS, $n=5$). 5 real ultrasound phantoms: 2 in-vivo carotid (cross-sectional and longitudinal) from the PICMUS challenge [15], 1 experimental contrast phantom, 1 resolution phantom, and 1 tissue-mimicking phantom (DeepUS). Compound plane-wave DAS beamforming, 75 angles. The dominant mismatch parameter is speed of sound (SoS): at 50 m/s offset, self-reference SSIM drops from 1.0 to 0.83 ± 0.04 ; after autonomous SoS calibration (grid search over 1400–1700 m/s), SSIM recovers to 0.98 ± 0.01 ($\rho_{\text{recov}} \approx 0.95$). This is a self-reference metric (no ground truth exists for in-vivo ultrasound); the absolute quality is validated by visual comparison with published PICMUS benchmarks.

E-ptychography (SrTiO₃ 4D-STEM). Real 4D scanning transmission electron microscopy data: SrTiO₃ [001] acquired with a Medipix3RX detector at 300 kV (Zenodo 5113449 [16]). Integrated centre-of-mass (iCoM) phase retrieval on a 64×64 scan subset. Probe-position jitter is the dominant mismatch: at 2 px standard deviation, self-reference PSNR drops from ∞ to 24.1 dB; oracle position correction recovers to 61.9 dB with phase correlation 0.9999 ($\rho_{\text{recov}} = 1.00$). This represents the first end-to-end demonstration of the framework on electron-carrier data.

Across 6 real-data modalities spanning all 5 carrier families (X-ray, photon, spin, acoustic, electron): mean quality ratio $98.1 \pm 0.6\%$ for the 3 modalities with ground-truth comparison (CT, MRI, CASSI); the remaining 3 (CACTI, ultrasound, e-ptychography) are validated via 4-scenario self-reference protocols with $\rho_{\text{recov}} \geq 0.95$.

2.6 Agent Ablation Study

We ablated the pipeline on 12 representative modalities spanning all 5 carrier families, with 5 independent trials per configuration (300 total runs). Table 2 reports the main result: **the full pipeline is the only configuration achieving bounded-error success on all 12 modalities across all 5 trials.**

Removing $\mathcal{A}_J$ (Judge): 4/12 specifications pass compilation but are physically invalid (wrong carrier, missing modulation, incompatible noise model, unstable CFL). 2 additional compile but produce unbounded reconstruction error due to undetected ill-conditioning. The Judge is

Table 2. Ablation study (5 trials $\times$ 12 modalities = 300 runs). “Valid”: passes all compiler gates. “ $\leq \varepsilon$ ”: reconstruction within Theorem 2 error bound. No single agent is dispensable.

Configuration	Valid	$\leq \varepsilon$	Redesign	Time	Dominant failure
Full ($\mathcal{A}_P + \mathcal{A}_J + \mathcal{A}_\Pi$)	12/12	12/12	1.7 ± 0.6	4.8 min	—
No $\mathcal{A}_J$	8.2/12	6.4/12	0	2.9 min	Invalid spec, unbounded ε
No $\mathcal{A}_\Pi$	12/12	9.6/12	1.7 ± 0.6	3.4 min	Suboptimal parameters
No $\mathcal{A}_P$ (manual)	10.4/12	10.4/12	0	52 min	Human error (14%)
Template only	—	3/12	0	0.5 min	No coverage (75%)

necessary for both safety and correctness.

Removing $\mathcal{A}_\Pi$: all 12 pass validation but 2–3 produce suboptimal reconstruction per trial (e.g., too few CT projections for target PSNR). Mean PSNR drop: 2.1 dB.

Removing $\mathcal{A}_P$: human experts write `spec.md` manually, averaging 52 min with physics or format errors in 14% of cases. The Plan Agent reduces specification time by $>10\times$.

Template only: covers 3/12 modalities (CT, MRI, lensless); all others fail.

2.7 Judge and Performance Agents

Judge rejection accuracy ($n=150$: 75 valid, 75 ill-posed, stratified by error category). Precision: 96.0% (95% CI: [91%, 99%]). Recall: 96.0%. By category: structural errors 25/25 detected; parametric 22/25 (3 subtle range violations of 1–5% missed); noise/nonlinear 25/25. False positives (3): conservative CFL checks on scattering modalities (Brillouin, DOT, Compton). Mean redesign rounds on rejection: 1.4 (max 3).

Performance prediction (36 modalities, 1,800 predictions). Overall MAE: 1.8 dB (95% CI: [1.5, 2.1]), $R^2 = 0.91$ ([0.87, 0.94]). By modality group: well-characterized ($n=20$) MAE 0.9 dB; moderate ($n=10$) 2.1 dB; scattering-dominated ($n=6$) 3.8 dB. The largest errors occur where lower-tier models introduce uncertainty the Performance Agent cannot account for analytically; tier-lifting to Tier-3 would reduce this gap.

2.8 Expert Comparison

Table 3. Agent vs. expert manual design. Bold = real measured data. Expert times from specification to first successful reconstruction, cross-validated against [4, 10]; remaining estimated from published timelines (see Methods).

Modality	PSNR (dB)		Setup		Quality	LoC
	Agent	Expert	Agent	Expert		
CT (real)	31.7	32.1	25 min	7 d	98.8%	12 vs 340
MRI (real)	31.7	32.4	20 min	5 d	97.8%	14 vs 520
CASSI (real)	29.5	30.2	18 min	4 d	97.7%	11 vs 280
SIM	32.1	32.8	15 min	3 d	97.9%	10 vs 210
OCT	30.5	31.2	22 min	6 d	97.8%	13 vs 450
DOT	24.1	25.3	30 min	10 d	95.3%	15 vs 680

Mean quality ratio: $97.6 \pm 1.1\%$. Quality gap largest for DOT (95.3%), where Tier-2 operation contributes the largest $\varepsilon_{\text{unmod}}$; this gap closes under tier-lifting (Table 5). Agent specifications: 10–15 lines; expert code: 210–680 lines.

2.9 Cross-Tier Hybrid Reconstruction

Each FPB primitive is implemented at all four fidelity tiers (Table 4), with dedicated adapter classes providing tier-specific validation. A `TierPolicy` automatically selects the tier based on compute budget and modality requirements.

Table 4. Four-tier fidelity hierarchy. All 11 primitives are implemented at every tier.

Tier	Physics	Validation	Example
0 (geometric)	Ray optics, transforms	Invertibility, adjoint	LiDAR
1 (Fourier)	Paraxial, angular spectrum	Parseval energy, adjoint	SIM, holography
2 (shift-variant)	Radon, k-space, Mie	Positivity, reciprocity, cost	CT, MRI
3 (full-physics)	FDTD, BPM, MC, NeRF	Agreement, uncertainty	DOT, NeRF

When a modality benefits from combining fidelity levels, the *tier-lifting protocol* provides hybrid reconstruction via a Hybrid Proximal-Gradient (HPG) method (Proposition 3, Methods). We validated tier-lifting on two cross-tier combinations (Table 5): DOT (Tier-2 P_1 diffusion $\rightarrow$ Tier-3 P_3 radiative transfer) and coherent imaging (Tier-1 angular spectrum $\rightarrow$ Tier-3 FDTD). In both cases, $\varepsilon_{\text{unmod}}$ reduces to machine precision ($< 10^{-12}$).

Table 5. Tier-lifting validation on two cross-tier combinations. Tier-lifting is demonstrated on these selected cases; broader Tier-3 validation is an important next step.

Domain	Low tier	High tier	Correction	$\varepsilon_{\text{unmod}}$	$\varepsilon_{\text{lift}}$
DOT	Tier-2 (P_1)	Tier-3 (P_3 RT)	Higher-order moments	7.8%	$< 10^{-12}$
Coherent	Tier-1 (ang. spec.)	Tier-3 (FDTD)	Evanescent + scatter	147%	$< 10^{-12}$

The tier-lifting protocol preserves the FPB structure: the same 11 primitives, the same DAG, the same `spec.md`—only the fidelity of each primitive changes. These two demonstrations show that the framework can in principle span all four tiers, but we emphasise that broader empirical validation across additional Tier-3 modalities is needed before claiming general Tier-3 coverage.

3 Discussion

Limitations. (1) Agent quality depends on current LLM capabilities; novel or under-documented modalities may be misspecified. (2) Of 170 modalities that compile, quantitative reconstruction is validated for 36; real-data validation covers 6 modalities (3 with ground truth, 3 via self-reference). Compilation alone does not guarantee reconstruction quality. (3) Expert comparison relies partly on published timeline estimates; prospective head-to-head studies would strengthen the claims. (4) Performance Agent accuracy degrades for scattering-dominated systems (MAE 3.8 dB vs. 0.9 dB for well-characterised modalities). (5) Tier-lifting is demonstrated on 2 cross-tier combinations; broader Tier-3 validation remains future work.

The specification as exchange standard. Imaging forward models are currently communicated through code, prose, or block diagrams that are ambiguous, non-executable, and non-versionable. `spec.md` addresses this by decoupling the *what* from the *how* in a structured 7-field document, enabling non-expert scientists to design imaging systems via natural language.

Formal verification versus ad hoc design. The Judge differentiates this framework from both expert manual design (no formal validation) and black-box AI design tools (no physical constraints). Its 96% rejection precision on 150 test specifications indicates robust detection of structural, parametric, and noise-model errors.

Bridging the reality gap. Theorem 2 provides a diagnostic protocol: each ε term maps to

a corrective action. In practice, $\varepsilon_{\text{spec}} = 0$ when the compiler certifies the design, $\varepsilon_{\text{trans}} = 0$ for 86% of validated modalities, and $\varepsilon_{\text{unmod}} = 0$ at Tier-3. The dominant residual is $\varepsilon_{\text{param}}$, reducible by calibration. The 4-scenario recovery protocol [2] confirms near-complete recovery for low-dimensional mismatch (CT, MRI) and 85–100% for multi-parameter systems (CASSI, CACTI, SPC).

Relation to existing work. AlphaFold [5] and GNoME [6] apply AI to specific scientific problems at scale; SigPy [10], ODL [11], and DeepInverse [9] provide operator libraries. Our framework provides the *design* layer that precedes operator instantiation, with formal error guarantees absent from both data-driven discovery and existing libraries.

Outlook. Extending real-data validation to ultrasound, OCT, and PET; validating tier-lifting across more Tier-3 modalities; improving Performance Agent calibration for scattering systems; and establishing a community `spec.md` benchmark are concrete next steps.

4Methods

4.1Proof of Theorem 2

Assumptions.

A1. Coercivity. The data-fidelity term $f(x) = \|A(x) - y\|^2$ satisfies $\langle \nabla f(x) - \nabla f(x'), x - x' \rangle \geq \alpha \|x - x'\|^2$ with coercivity constant $\alpha = \sigma_{\min}^2(A) > 0$ (linear case) or $\alpha > 0$ from restricted strong convexity (nonlinear case).

A2. Regularisation. $R(x)$ is proper, convex, lower-semicontinuous, and $\lambda > 0$. The composite objective $F(x) = f(x) + \lambda R(x)$ is μ -strongly convex with $\mu = \alpha + \lambda$.

A3. Lipschitz parameters. Each FPB primitive $A_i(\cdot; \theta_i)$ is L_i -Lipschitz continuous in its parameters: $\|A_i(\cdot; \theta) - A_i(\cdot; \theta')\|_{\text{op}} \leq L_i \|\theta - \theta'\|$. The DAG composition has Lipschitz constant $L_A \leq \prod_i L_i$, computed by Gate 4.

A4. Bounded signal. $\|x^*\| \leq B$ for known bound B .

Step 1: Operator decomposition. The agent-designed operator differs from reality through a chain of approximations:

$$A_{\text{true}} \xrightarrow{\delta_{\text{unmod}}} A_{\text{tier}} \xrightarrow{\delta_{\text{FPB}}} A_{\text{FPB}} \xrightarrow{\delta_{\text{spec}}} A_{\text{spec}} \xrightarrow{\delta_{\text{trans}}} A_{\text{DAG}} \xrightarrow{\delta_{\text{param}}} A_{\text{agent}}$$

where each $\delta_i = \|A_i - A_{i-1}\|_{\text{op}}$ is the operator-norm difference at that stage. By the triangle inequality:

$$\|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}} \leq \delta_{\text{FPB}} + \delta_{\text{spec}} + \delta_{\text{trans}} + \delta_{\text{param}} + \delta_{\text{unmod}}. \quad (2)$$

Step 2: Reconstruction perturbation. Define $x_{\text{oracle}} = \arg \min_x \|A_{\text{true}}(x) - y\|^2 + \lambda R(x)$ (oracle reconstruction using A_{true}). Define $\hat{x} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda R(x)$ (agent reconstruction). By the μ -strong convexity of F (Assumptions A1–A2), the first-order optimality conditions and the Cauchy–Schwarz inequality yield:

$$\|\hat{x} - x_{\text{oracle}}\| \leq \frac{2B}{\mu} \|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}} = \frac{2B}{\alpha + \lambda} \|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}}. \quad (3)$$

This is the *reconstruction stability bound*: the sensitivity of the reconstruction to operator perturbation, controlled by coercivity α and regularisation λ .

Step 3: Define effective error terms. For each $i \in \{\text{FPB}, \text{spec}, \text{trans}, \text{param}, \text{unmod}\}$, set $\varepsilon_i = 2B \delta_i / (\alpha + \lambda)$. Then:

$$\|\hat{x} - x_{\text{oracle}}\| \leq \varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}. \quad (4)$$

Step 4: Total error. $\|\hat{x} - x^*\| \leq \|\hat{x} - x_{\text{oracle}}\| + \|x_{\text{oracle}} - x^*\|$. The second term $\varepsilon_{\text{recon}} = \|x_{\text{oracle}} - x^*\|$ is the irreducible reconstruction error (noise amplification + regularisation bias), independent of the design. Combining:

$$\|\hat{x} - x^*\| \leq (\varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}) + \varepsilon_{\text{recon}}.$$

Step 5: Bounding each term.

- (a) $\delta_{\text{FPB}} < 0.01$ by the FPB basis theorem [2]. Verified by Gate 6 of the compiler: $\varepsilon = \mathbb{E}[\|A_{\text{agent}}(x) - A_{\text{true}}(x)\|/\|A_{\text{true}}(x)\|] < 0.01$ over 5 random test vectors. Hence $\varepsilon_{\text{FPB}} < 0.01 \cdot 2B/(\alpha + \lambda)$.
- (b) $\delta_{\text{spec}} = 0$ when all 6 compiler gates pass. *Condition:* Gates 1–5 jointly verify that the `spec.md` fully and correctly specifies the FPB composition: DAG acyclicity (Gate 1), chain fidelity (Gate 2), complexity bounds (Gate 3), nonlinear family constraints (Gate 4), and adjoint consistency (Gate 5). When all pass, $A_{\text{spec}} \equiv A_{\text{FPB}}$.
- (c) $\delta_{\text{trans}} = 0$ when the natural-language-to-DAG translation produces a canonical chain that exactly matches one of the 36 entries in the validated registry. *Condition:* Gate 2 (chain fidelity) returns exact match. This holds for 31/36 (86%) of validated modalities.
- (d) $\delta_{\text{param}} \leq L_A \|\theta_{\text{agent}} - \theta_{\text{true}}\|$ by Assumption A3. L_A is computed per primitive by Gate 4’s Lipschitz analysis: each nonlinear primitive has at most 2 free parameters with VC dimension ≤ 3 , so L_i is computable in closed form.
- (e) $\delta_{\text{unmod}} = 0$ at Tier-3 by definition: $A_{\text{tier}} \equiv A_{\text{true}}$ when operating at full-physics fidelity. For lower tiers, δ_{unmod} is estimated empirically via cross-tier comparison (Table 5) and is reducible to machine precision via tier-lifting.

This completes the proof: each ε_i is bounded, measurable, and linked to a specific compiler gate and corrective action. $\square$

4.2Tier-Lifting Protocol

Proposition 3 (Tier-Lifting). *Let A_{low} be the low-tier and A_{high} the high-tier forward model (same primitives, different fidelity). Define $\Gamma(x) = A_{\text{high}}(x) - A_{\text{low}}(x)$. Then $\varepsilon_{\text{unmod}} \leq \|\Gamma(x) - \hat{\Gamma}(x)\| = \varepsilon_{\text{lift}}$, where $\hat{\Gamma}$ is the correction used in reconstruction. In full-correction mode ($\hat{\Gamma} = \Gamma$), $\varepsilon_{\text{lift}} = 0$.*

The Hybrid Proximal-Gradient (HPG) loop:

$$x_{k+1} = \text{prox}_{\lambda R}(x_k - \gamma A_{\text{low}}^T(A_{\text{high}}(x_k) - y))$$

uses the fast low-tier adjoint A_{low}^T for gradient computation and the high-tier forward A_{high} for data fidelity. Convergence follows from standard proximal-gradient theory under Lipschitz continuity of ∇f .

4.36-Gate Compiler

Gate 1 (DAG). Compiled into a typed `OperatorGraphSpec` (Pydantic v2). Acyclicity verified via Kahn’s algorithm, $O(V+E)$.

Gate 2 (Chain). Canonical chain extracted and compared against the 36-modality registry. Match is exact or within known equivalences (Σ merged into D).

Gate 3 (Bounds). $N \leq 20$ nodes, $D \leq 10$ depth.

Gate 4 (Nonlinear). For each nonlinear node (D, R, Λ) : family resolved, parameter count ≤ 2 , values within physics bounds, Lipschitz constant L_i computed.

Gate 5 (Adjoint). Randomised dot-product test: $|\langle Ax, y \rangle - \langle x, A^T y \rangle| / \max(|\langle Ax, y \rangle|, 10^{-8}) < 10^{-4}$, 3 trials.

Gate 6 (Error). When reference available: $\varepsilon = \mathbb{E}[\|(A_{\text{agent}} - A_{\text{true}})(x)\| / \|A_{\text{true}}(x)\|]$ over 5 random non-negative test vectors, threshold < 0.01 .

4.44-Scenario Recovery Protocol

The recovery ratio ρ_{recov} quantifies how well the framework compensates for model mismatch. Following [2], we define four scenarios validated on 6 representative modalities spanning 4 carrier families (CASSI, CACTI, SPC: photon; CT: X-ray; electron ptychography: electron; MRI: spin):

$$\text{Sc. I (oracle): } y = A_{\text{true}}(x) + n; \hat{x}_{\text{I}} = \arg \min_x \|A_{\text{true}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (5)$$

$$\text{Sc. II (mismatch): } y = A_{\text{true}}(x) + n; \hat{x}_{\text{II}} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (6)$$

$$\text{Sc. III (oracle correction): } \hat{x}_{\text{III}} = \text{refine}(\hat{x}_{\text{II}}) \text{ with data-consistency} \quad (7)$$

$$\text{Sc. IV (autonomous calibration): } \hat{x}_{\text{IV}} = \arg \min_x \|A_{\text{agent}}^{\text{cal}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (8)$$

The recovery ratio $\rho_{\text{recov}} = (\text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}}) / (\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ measures autonomous calibration effectiveness relative to the oracle correction.

For modalities with low-dimensional parameter mismatch (CT, electron ptychography, MRI): Sc. IV $\approx$ Sc. III $\approx$ Sc. I, confirming near-complete recovery ($\rho_{\text{recov}} \approx 1.0$). For multi-parameter mismatch: CASSI 85%, CACTI 100%, SPC 86%. Across all 36 modalities: mean $\rho_{\text{recov}} = 0.85 \pm 0.07$.

4.5Expert Time Methodology

Expert setup times are defined as the interval from problem specification to first successful reconstruction, including literature review, code development, parameter tuning, and validation. For CT and MRI, times are cross-validated against published development reports [4, 10] and confirmed by domain experts. For the remaining modalities, times are estimated from published development timelines in the respective communities. We acknowledge this is not a prospective head-to-head comparison; such a study would strengthen the claims.

4.6Implementation

The framework is implemented in Python (Physics World Model platform). Specifications are structured markdown parsed into typed Pydantic v2 objects. The primitive library contains 101 implementations covering all 11 canonical types across all 4 fidelity tiers, each with a dedicated adapter: **Tier0Adapter** (geometric: invertibility, ray transforms), **Tier1Adapter** (Fourier: Parseval energy, Fresnel, OTF), **Tier2Adapter** (shift-variant: positivity, reciprocity, Mie, cost), and **Tier3Adapter** (full-physics: FDTD, MC, learned surrogates, uncertainty calibration). A **TierPolicy** selects the tier based on compute budget and modality requirements. Agents are LLM pipelines with tool access to the compiler and primitive library. Mean compilation time: 0.35 ms. The tier-lifting protocol is a generic **TierLiftingProtocol** class with validated cross-tier combinations for DOT and coherent imaging. Source code: https://github.com/integritynoble/Physics_World_Model.

Extended Data

Data Availability

The LoDoPaB-CT dataset is publicly available [3]. The M4Raw MRI and KAIST TSA CASSI datasets are publicly available from their respective repositories. The PICMUS ultrasound data is from the IEEE IUS challenge [15]. The SrTiO₃ 4D-STEM data is publicly available on Zenodo [16]. The CACTI real video data is from the project repository. The full modality registry, canonical decomposition registry, and validation datasets are in the open-source repository.

Code Availability

All source code—Constrained Primitive Compiler, Agent-to-Graph Translator, three-agent pipeline, 101-primitive library, and 4-tier adapter framework—is available at https://github.com/integritynoble/Physics_World_Model under the MIT licence.

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Table 6. Extended Data Table 1: Modality registry (36 validated canonical decompositions).
Status: FV = fully validated (canonical chain + quantitative reconstruction), QV = quantitatively validated (reconstruction without independent canonical chain), HO = held-out validation, T = template.
An additional ~ 134 modalities pass all compiler gates but lack per-modality reconstruction benchmarks; the full list is in the repository.

Modality	Domain	Carrier	Tier	Chain	Status	Real?	Dom. ε
<i>X-ray</i>							
CT	Medical	X-ray	1–2	$\Pi \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
CBCT	Medical	X-ray	2	$\Pi \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
CT (polychromatic)	Medical	X-ray	2	$\Pi \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Phase contrast	Medical	X-ray	1–2	$\Pi \rightarrow P \rightarrow M \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
PET	Nuclear	X-ray	2	$\Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
SPECT	Nuclear	X-ray	2	$\Pi \rightarrow S \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
DEXA	Medical	X-ray	2	$M \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Compton	Nuclear	X-ray	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
<i>Optical photons</i>							
CASSI	Spectral	Photon	2	$M \rightarrow W \rightarrow \Sigma \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
CACTI	Video	Photon	2	$M \rightarrow \Sigma \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
SPC	Imaging	Photon	2	$M \rightarrow \Sigma \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
OCT	Ophthal.	Photon	1–2	$P+P \rightarrow \Sigma \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
SIM	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
STED	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
PALM/STORM	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
TIRF	Micro.	Photon	1	$P \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Ptychography	Micro.	Photon	1–2	$M \rightarrow P \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
Lensless	Imaging	Photon	1	$C \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
DOT	Diffuse	Photon	2–3	$M \rightarrow R \rightarrow P \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Ghost imaging	Quantum	Photon	2	$M \rightarrow \Sigma \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
THz-TDS	Spectro.	Photon	1	$C \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
Light field	3D	Photon	1	$M \rightarrow C \rightarrow S \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Raman	Spectro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Fluorescence	Micro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Fluorescence (sat.)	Micro.	Photon	2	$M \rightarrow R \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Brillouin	Spectro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
<i>Spin</i>							
MRI	Medical	Spin	1–2	$M \rightarrow F \rightarrow S \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
MRI (phase)	Medical	Spin	2	$M \rightarrow F \rightarrow S \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
<i>Acoustic</i>							
Ultrasound	Medical	Acoustic	2	$P \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
Doppler US	Medical	Acoustic	2	$P \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Elastography	Medical	Acoustic	2	$P \rightarrow P \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Photoacoustic	Medical	Acoustic	2	$M \rightarrow P \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
<i>Electron</i>							
SEM	Material	Electron	2	$M \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
TEM	Material	Electron	2	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
E-ptychography	Material	Electron	2	$M \rightarrow P \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
<i>Particle</i>							
Muon tomography	Security	Particle	2	$R \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Proton therapy	Medical	Particle	2	$\Lambda \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$

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